

Table 1: In silico characterization and prioritization of HPV16 E6/E7-derived wild-type epitopes and altered peptide ligand (APL) candidates restricted by HLA-A*02:01.

Protein	Epitope ID	Peptide Sequence	WT Sequence	Substitution	HLA	Position	Immunogenicity	Antigenicity	Toxicity	Presentation Score	%Rank (EL)	Affinity (nM)	Binding Level
E6	E6wt	TIHDIILECV	—	—	A*02:01	29–38	−0.008	1.00	Non-Toxin	0.286	0.990	329.33	WB
E6	E6apl	Y LHDIILECV	TIHDIILECV	T1→Y; I2→L	A*02:01	29–38	0.111	1.00	Non-Toxin	0.873	0.024	6.21	SB
E7	E7wt1	LLMGTLGIV	—	—	A*02:01	82–90	−0.008	0.66	Non-Toxin	0.238	1.258	12.59	WB
E7	E7apl1	Y LMGTLGIV	LLMGTLGIV	L1→Y	A*02:01	82–90	0.335	0.66	Non-Toxin	0.582	0.298	4.24	SB
E7	E7wt2	YMLDLQPETT	—	—	A*02:01	11–20	−0.008	0.66	Non-Toxin	0.164	1.442	207.32	WB
E7	E7apl2	YMLDLQPET V	YMLDLQPETT	T10→V	A*02:01	11–20	0.335	0.66	Non-Toxin	0.873	0.063	6.63	SB
E7	E7apl3	YLL DLQPET V	YMLDLQPETT	M2→L; T10→V	A*02:01	11–20	0.111	0.66	Non-Toxin	0.947	0.025	6.14	SB

Note: Mutated residues in APL sequences are highlighted in bold. **APL**, altered peptide ligand; **WT**, wild-type; **SB**, strong binder; **WB**, weak binder. Immunogenicity scores were predicted using the IEDB Class I Immunogenicity Predictor. Antigenicity was evaluated using VaxiJen v3.0, and toxicity was assessed using ToxinPred. Peptide presentation scores, %Rank values, binding affinities, and binding classifications were predicted using NetMHCpan-4.1. Peptides with %Rank < 0.5% were classified as strong binders (SB), whereas peptides with %Rank between 0.5% and 2.0% were classified as weak binders (WB).